

a) FIFO

```
#include <stdio.h>

int main() {
    int n, frames;

    printf("Enter number of pages in reference string: \n");
    scanf("%d", &n);

    int ref[n];

    printf("Enter the reference string: \n");
    for(int i = 0; i < n; i++) {
        scanf("%d", &ref[i]);
    }

    printf("Enter number of frames: ");
    scanf("%d", &frames);

    int frame[frames];
    int i, j;

    for(i = 0; i < frames; i++) {
        frame[i] = -1;
    }

    int pageFaults = 0, pageHits = 0;
    int index = 0;

    for(i = 0; i < n; i++) {
        int found = 0;

        for(j = 0; j < frames; j++) {
            if(frame[j] == ref[i]) {
                found = 1;
                pageHits++;
                break;
            }
        }

        if(!found) {
            frame[index] = ref[i];
            index = (index + 1) % frames;
            pageFaults++;
        }
    }

    printf("\n\nTotal Page Faults: %d\n", pageFaults);
    printf("Total Page Hits: %d\n", pageHits);

    return 0;
}
```

Custom Test

Success

Input:

7
1 3 0 3 5 6 3
3

Expected Output :

Enter number of pages in reference string:
Enter the reference string:
Enter number of frames:

Total Page Faults: 6
Total Page Hits: 1

Output:

Enter number of pages in reference string:
Enter the reference string:
Enter number of frames:

Total Page Faults: 6
Total Page Hits: 1

b) LRU

```
#include <stdio.h>

int main() {
    int pages, frames;

    printf("Enter number of pages in reference string: \n");
    scanf("%d", &pages);

    int ref[pages];

    printf("Enter the reference string: \n");
    for(int i = 0; i < pages; i++) {
        scanf("%d", &ref[i]);
    }

    printf("Enter number of frames: \n");
    scanf("%d", &frames);

    int frame[frames], time[frames];
    int faults = 0, hits = 0, counter = 0;

    // Initialize frames
    for(int i = 0; i < frames; i++) {
        frame[i] = -1;
        time[i] = -1;
    }

    for(int i = 0; i < pages; i++) {
        int found = 0;

        // Check Hit
        for(int j = 0; j < frames; j++) {
            if(frame[j] == ref[i]) {
                hits++;
                counter++;
                time[j] = counter;
                found = 1;
                break;
            }
        }

        // Fault
        if(!found) {
            int pos = -1;

            // Empty frame available
            for(int j = 0; j < frames; j++) {
                if(frame[j] == -1) {
                    pos = j;
                    break;
                }
            }

            // If no empty frame → find LRU
            if(pos == -1) {
                pos = 0;
                for(int j = 1; j < frames; j++) {
                    if(time[j] < time[pos]) {
                        pos = j;
                    }
                }
            }

            frame[pos] = ref[i];
            counter++;
            time[pos] = counter;
        }
    }
}
```

Custom Test

Success

Input:

Expected Output :

5
1 2 3 4 5
5

Enter number of pages in reference string:
Enter the reference string:
Enter number of frames:

Total Page Faults: 5
Total Page Hits: 0

Output:

Enter number of pages in reference string:
Enter the reference string:
Enter number of frames:

Total Page Faults: 5
Total Page Hits: 0